

Phillip Marlowe

FPGA/ASIC RTL and Physical Design Graduate

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Education

University of California, Santa Cruz

M.S. in Computer Science & Engineering

December 2025

B.S. in Computer Engineering conc. Digital Hardware

June 2023

Member of Engineering Honor Society Tau Beta Pi: Ca Alpha Delta Chapter

June 2022

Professional Experience

Hardware Systems Collective - UC Santa Cruz

Sep 2023 – Dec 2025

Graduate Student Researcher

- Produced a semi-generalized physical design flow to generate eFPGA fabric parasitics for use in observing Bias Temperature Instability (BTI), a form of transistor degradation, on particular nodes and routes.
- Created a SPICE test environment to observe BTI effects on metastability in D-FlipFlops and plotted clock-to-q delay to experimentally determine setup and hold times as well as point of failure.

Baskin Engineering Lab Support - UC Santa Cruz

May 2022 – Dec 2025

Student Technician

- Supported 20+ instructional labs by verifying, moving, and securing test & measurement equipment, as well as sourcing supplies to ensure that each lab was safe and functional for instruction.
- Assisted students and faculty by reconciling equipment checkouts, assembling and distributing instructional kits, operating specialized tools (3D printers, PCB mill, laser cutters), and producing standard operating procedures.
- Maintained lab workstations by installing and updating specialized software among enabling remote use, creating and deploying master images, and resolving hardware and software issues.
- Operated AV systems for Baskin Engineering event spaces, supported presenters, and resolved live issues.
- Managed stockroom, poster, and receiving workflows by tracking inventory, sourcing required materials, overseeing poster printing and documentation, and supporting precise shipping and receiving operations.

Capstone Engineering Design Project - UC Santa Cruz

Jan 2023 – Jun 2023

Electrical & Mechanical Hardware Systems Engineer

- Collaborated on a six-person team to build a hands-free following utility wagon by integrating electrical and mechanical subsystems to assist workers and older adults with carrying physical loads.
- Designed wiring and power distribution for motors, microcontrollers, and sensors, enabling reliable communication and consistent power across the system.
- Selected and sourced materials to assemble circuits; tested the prototype to verify proper functionality.

Engineering Projects

HSC TDC ASICs with varying delay lines - Hardware Systems Collective

Apr 2024

- Produced multiple delay line elements' RTL for use within Time to Digital Converters (TDCs)
- Produced Synopsis Design Constraint (SDC) file to constrain location and size of delay line standard cells
- Pushed multiple TDC designs with produced delay line elements through openROAD physical design flow to produce ASICs taped out in the TinyTapeout 6 shuttle using the Skywater130 PDK.

HLS Leaky Integrate and Fire Neuron ASIC - ECE 210: Brain-Inspired Machine Learning

Nov 2023

- Used Python based High Level Synthesis (HLS) tool to generate RTL for a Leaky Integrate and Fire Neuron (LIFN)
- Pushed generated LIFN RTL through openROAD physical design flow to produce ASIC taped out in the TinyTapeout 5 shuttle using the Skywater130 PDK.

Ground Bounce PCB - ECE 173: High-Speed Digital Design

Jun 2023

- Designed and fabricated a custom printed circuit board to experimentally observe ground bounce.
- Measured propagation delays and signal distortion with an oscilloscope to evaluate switching noise, validating results through RC timing and frequency testing.

Technical Skills

- **Software/Design:** 3 years of FPGA/ASIC design experience, Cadence ASIC Design Tools, Cadence PCB Design Tools, Xilinx Vivado, Synopsys VCS, open-source ASIC/FPGA design tools, SolidWorks, Windows support, Linux support, Docker, Git, LaTeX, Markdown, web development
- **Programming Languages:** SystemVerilog (including various HLS), C/C++, Python, Java, JavaScript, TCL, Bash, SPICE analog simulation, Makefile automation, ISAs (RISC-V, x86, MIPS)
- **Hardware Experience:** Xilinx and Lattice FPGAs, Arduino, Raspberry Pi, Microchip MCUs, Cypress PSoC 6, STM32
- **Fabrication:** 2-layer PCB milling, surface mount and through-hole soldering, PLA/SLA printing, laser cutting, electronics diagnosis & repair